

# Assignment Two - Machine Learning: Details

## Overview

### Task description

Your task is to use machine learning to train a classifier capable of distinguishing different classes of flowers in an image, and to evaluate this classifier's performance. For this we will be using deep learning and a technique called Transfer Learning. The assignment also includes an individual reflection on client delivery and ethical considerations.

### Conditions

|                    |                                  |
|--------------------|----------------------------------|
| Weighting          | 30%                              |
| Assessment Type    | Project (applied)                |
| Group / Individual | Group (20%) and Individual (10%) |

### Unit Learning outcomes measured

- 2. Analyse machine learning problems and formalise them in the language/framework of relevant AI methods.
- 3. Build intelligent artefacts using AI software engineering skills.
- 4. Evaluate empirically different solutions and make fully justified recommendations.
- 5. Collaborate with others in a team environment to deliver an outcome for a client.
- 6. Communicate professionally in writing to produce a report for a client.

## What you need to do

Further details, including required content for the assessment can be found in the instructions document below.

### Resources available to complete the task

- **Detailed instructions** and background information: [CAB320-Assignment2-ML-Instructions.pdf \(https://canvas.qut.edu.au/courses/23773/files/8288312?wrap=1\)](https://canvas.qut.edu.au/courses/23773/files/8288312?wrap=1). 
- **Dataset** of images provided (zip file to download and unzip: [small flower dataset.zip \(https://canvas.qut.edu.au/courses/23773/files/7631675?wrap=1\)](https://canvas.qut.edu.au/courses/23773/files/7631675?wrap=1).  [https://canvas.qut.edu.au/courses/23773/files/7631675/download?download\\_frd=1](https://canvas.qut.edu.au/courses/23773/files/7631675/download?download_frd=1) ). The dataset is composed of 4 directories, one per class of flower (daisy, dandelion, roses, sunflowers, tulips). Each directory contains images featuring the corresponding class of flower.
- **Python Jupyter Notebook template** file to download: [CAB320-Assignment2-Learning.ipynb \(https://canvas.qut.edu.au/courses/23773/files/7631719?wrap=1\)](https://canvas.qut.edu.au/courses/23773/files/7631719?wrap=1).  [https://canvas.qut.edu.au/courses/23773/files/7631719/download?download\\_frd=1](https://canvas.qut.edu.au/courses/23773/files/7631719/download?download_frd=1)

## What you need to submit

### Submission requirements

You need to submit **two files** + an evaluation of your group members:

- 1. **One Jupyter Notebook file per group (or one Python script, i.e. one file)** with your code and your reporting (see Group Tasks section in the instructions)
- 2. **One individual report per student, in pdf format, strictly limited to 3 pages** (anything beyond 3 pages will be ignored by the marker). This does not include the cover page if you have one (not necessary).
- 3. An **evaluation of your group members** (including yourself), using the evaluation provided (see item 'Assignment Two - Group Member Evaluation'). This is required if a moderation may be needed in your group, optional otherwise. (NB: No moderation of the group can be requested if this has not been completed by all team members).

### Group members:

- Don't forget to **list all the members of your team in the Notebook**
- Only **one person in your team should submit the assignment**. The others should always check that the submission was completed and the file submitted is the right one.
- The team size is limited to 3. Teams of size 2 are accepted. However, the tasks to be completed remain the same.
- Please complete the Group Member Evaluation (coming soon), especially if there were issues in the group work during the preparation of the assignment.



### Generative AI tools may be used ethically and responsibly:

You may use generative artificial intelligence (GenAI) tools to prepare for, generate and refine content for this assessment task.

AI-generated content may be inaccurate, unreliable, or biased. It is your responsibility to critically evaluate any information you use.

You must clearly acknowledge and appropriately reference any AI-generated content following the guidance in [Cite | Write \(https://www.citewrite.qut.edu.au/\)](https://www.citewrite.qut.edu.au/cite/qutcite.html#apa-internet-ai). ([APA \(https://www.citewrite.qut.edu.au/cite/qutcite.html#apa-internet-ai\)](https://www.citewrite.qut.edu.au/cite/qutcite.html#apa-internet-ai), [Harvard \(https://www.citewrite.qut.edu.au/cite/qutcite.html#harvard-internet-ai\)](https://www.citewrite.qut.edu.au/cite/qutcite.html#harvard-internet-ai), [AGLC \(https://www.citewrite.qut.edu.au/cite/qutcite.html#aglc-internet-ai\)](https://www.citewrite.qut.edu.au/cite/qutcite.html#aglc-internet-ai), or [Vancouver \(https://www.citewrite.qut.edu.au/cite/qutcite.html#vancouver-internet-ai\)](https://www.citewrite.qut.edu.au/cite/qutcite.html#vancouver-internet-ai)). You may be asked to demonstrate authorship of your assessment. Find out more on [keeping good records to authenticate authorship \(https://qutvirtual4.qut.edu.au/documents/d/student/keeping-good-records-to-authenticate-authorship\)](https://qutvirtual4.qut.edu.au/documents/d/student/keeping-good-records-to-authenticate-authorship).